gkwdist: An R Package for the Generalized Kumaraswamy Distribution Family

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Summary

`gkwdist` provides the distribution layer of the Generalized Kumaraswamy (GKw) family (Carrasco et al. 2010) in R: densities, distribution and quantile functions, random generation, and exact analytical log-likelihoods, score vectors and Hessian matrices for the five-parameter GKw distribution and each of its seven nested sub-families. The analytical derivatives are what set it apart from existing implementations of these distributions. All routines are implemented in C++ through `RcppArmadillo` (Eddelbuettel and Sanderson 2014; Eddelbuettel and François 2011) and evaluated entirely in log space, which keeps them accurate for data concentrated near the boundaries of $(0, 1)$, where the naive algebra loses every significant digit. The analytical derivatives make gradient-based maximum likelihood estimation, observed-information standard errors and likelihood ratio tests available across the hierarchy without numerical differentiation.

Statement of Need

Bounded continuous data on $(0, 1)$, such as proportions, rates and indices, are ubiquitous, and the Beta (Ferrari and Cribari-Neto 2004; Smithson and Verkuilen 2006) and Kumaraswamy (Kumaraswamy 1980; Jones 2009) distributions are the standard tools for them. Both struggle with samples that are simultaneously asymmetric, heavy-tailed and concentrated near a boundary. The GKw family embeds both as special cases within a five-parameter hierarchy, so an analyst can start from a two-parameter model and adopt extra parameters only when a likelihood ratio test justifies them.

That workflow needs three things the existing ecosystem does not supply together: every member of the hierarchy under one interface, exact derivatives of each log-likelihood, and numerics that survive the boundary. The third is not a detail. For $x = 10^{-7}$ and $\alpha = 3.5$, the quantity $1 - x^\alpha$ rounds to exactly 1 in double precision, so the inner factors of the density collapse to zero and their logarithms are lost. Clamping them at a small constant instead of using a $\log(1 - e^u)$ formulation (Mächler 2012) yields log-likelihoods wrong by thousands of units, which silently corrupts AIC, BIC and every likelihood ratio test built on them.

The GKw Family

Let $X \sim \text{GKw}(\alpha, \beta, \gamma, \delta, \lambda)$ on $(0, 1)$, with $\alpha, \beta, \gamma, \lambda > 0$ and $\delta \geq 0$. Writing the three chained quantities

$$v = 1 - x^\alpha, \qquad w = 1 - v^\beta, \qquad z = 1 - w^\lambda,$$

the density and distribution functions are

$$f(x; \boldsymbol{\theta}) = \frac{\lambda\alpha\beta}{B(\gamma, \delta + 1)}\, x^{\alpha-1}\, v^{\beta-1}\, w^{\gamma\lambda-1}\, z^{\delta}, \qquad F(x; \boldsymbol{\theta}) = I_{w^\lambda}(\gamma, \delta + 1),$$

where $\boldsymbol{\theta} = (\alpha, \beta, \gamma, \delta, \lambda)^\top$, $B(\cdot, \cdot)$ is the beta function and $I_y(a, b)$ the regularized incomplete beta function. For a sample $x_1, \ldots, x_n$ the log-likelihood is

$$\ell(\boldsymbol{\theta}) = n \log \frac{\lambda\alpha\beta}{B(\gamma, \delta+1)} + \sum_{i=1}^{n} \Big[(\alpha-1)\log x_i + (\beta-1)\log v_i + (\gamma\lambda-1)\log w_i + \delta \log z_i\Big].$$

Every score and Hessian entry follows from differentiating these nested logarithms, so each can be written as a single exponentiated difference of log quantities, for example $\partial \log w / \partial \alpha = \beta \log(x) \exp(\alpha \log x + (\beta - 1)\log v - \log w)$. `gkwdist` evaluates $\log v$, $\log w$ and $\log z$ through one stable kernel and implements every derivative in that form, so the score and Hessian stay accurate wherever the log-likelihood itself does.

The family arises from the Kumaraswamy-G construction (Cordeiro and Castro 2011; Nadarajah et al. 2012), and seven sub-families follow from parameter constraints (Table 1). The McDonald case recovers the generalized beta of the first kind (McDonald 1984), and the Beta distribution appears at $\alpha = \beta = \lambda = 1$ as Beta$(\gamma, \delta + 1)$. The Kw, KKw and EKw members admit closed-form quantiles, so their random generation is direct inversion.

Table 1: GKw sub-family hierarchy.

Sub-family	Par.	Constraint	Closed quantile
GKw (`gkw`)	5	none	No
BKw (`bkw`)	4	$\lambda = 1$	No
KKw (`kkw`)	4	$\gamma = 1$	Yes
EKw (`ekw`)	3	$\gamma = 1, \delta = 0$	Yes
Mc (`mc`)	3	$\alpha = \beta = 1$	No
Kw (`kw`)	2	$\gamma = \delta = 0, \lambda = 1$	Yes
Beta (`beta_`)	2	$\alpha = \beta = \lambda = 1$	No

State of the Field

`extraDistr` (Wolodzko 2020) supplies basic Kumaraswamy density, distribution and quantile functions but no likelihood derivatives. `VGAM` (Yee 2015) offers Kumaraswamy regression via its `kumar()` family and `betareg` (Cribari-Neto and Zeileis 2010) covers Beta regression, but both are regression packages exposing no reusable distribution-level derivative interface. `gamlss.dist` (Rigby et al. 2019) includes Beta variants, and `unitquantreg` (Mazucheli et al. 2022) implements quantile regression for several unit-interval distributions. None covers the GKw hierarchy, and none exports analytical score and Hessian functions a user can pass to an optimiser.

`gkwdist` is deliberately the distribution layer only, and that split is not incidental. It was made during the review of `gkwreg` (Lopes and Bonat 2026) at the reviewers' request, to reduce that package's complexity and improve maintainability; `gkwreg` 2.0.0 moved its `d/p/q/r` functions here and now imports them, retaining regression with covariates and estimation through `TMB` (Kristensen et al. 2016). The separation leaves the densities and derivatives usable by anyone fitting a GKw model with `optim()`, `nlminb()`, or their own machinery. It also makes them independently testable, which is what made the componentwise audit described below possible.

Implementation and Validation

The package exports 49 functions following R's distribution conventions: `d/p/q/r` for each sub-family, plus `ll`, `gr` and `hs` for the negative log-likelihood, its gradient and its Hessian, and `gkwgetstartvalues()` for moment-based starting values. The theory vignette derives the score and observed information for each sub-family in full.

Because every sub-family is the GKw distribution with parameters fixed, each specialised routine has an exact reference: the general GKw routine evaluated at the constrained point. The test suite exploits this.

Every gradient component and every Hessian entry is compared individually against both that reference and Richardson extrapolation (Gilbert and Varadhan 2019), rather than through an aggregate norm, over grids including the degenerate values $\gamma = 1$, $\beta = 1$, $\lambda = 1$, $\delta = 0$ and samples with observations near zero. Those points matter: a mixed derivative such as $\partial^2\ell/\partial\alpha\,\partial\gamma$ loses its $(\gamma - 1)$ factor under differentiation and so does not vanish at $\gamma = 1$, precisely where a guarded implementation is most likely to be wrong. Across 720 configurations per family the maximum componentwise relative error is 4×10^{-8} against Richardson extrapolation and 10^{-13} against the GKw reference; log-likelihoods match an independent log-space reference to 2×10^{-12}. Densities are additionally checked to integrate to one and to reproduce base R for the Beta and closed-form Kumaraswamy cases.

At $n = 2 \times 10^4$ the analytical score is about 9 times faster than Richardson extrapolation and the analytical Hessian about 38 times faster, and supplying the analytical gradient roughly halves the time to convergence in full maximum likelihood fits. In a recovery study of 1,400 fits the optimiser reached a log-likelihood at least as high as the one at the data-generating parameter in essentially every replicate, while individual estimates of the four- and five-parameter members were often far from their true values. This illustrates the weak identifiability of the larger models directly, and it is why the package is built to make the parsimonious members easy to fit and compare.

Research Impact

`gkwdist` is the computational foundation of `gkwreg` (Lopes and Bonat 2026), published in this journal, which declares it among its imports and relies on it for every density, distribution and quantile evaluation. The package has been on CRAN since November 2025 and downloaded more than 4,400 times, currently around 160 times a month. Development has proceeded publicly since October 2025 across five tagged releases, with continuous integration, a changelog, contribution guidelines and an issue tracker.

AI Usage Disclosure

Claude (Anthropic) assisted with prose editing of the documentation and this manuscript, and with the numerical audit that compared each analytical derivative component against independent references. The distribution theory, the derivations of the score and Hessian, the C++ implementation and all design decisions are the author's own. Every AI-assisted output was reviewed and validated by the author, and all numerical claims reported here are reproducible from the scripts and test suite in the repository.

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